

Quick Check

Lesson 10: Synthesis and Application: Explaining the Farmer's Maize and Beans and Planning the Tree Nursery

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 10

A school in Machakos wants to grow 200 mango and grevillea seedlings from stem cuttings. Which idea would best help the cuttings form roots?

- A** Remove all the leaves and the cut surface so nothing is left to use water
- B** Keep the cuttings in full sun without water so they become strong
- C** Spread fertiliser on the cut end, because fertiliser makes roots form
- D** Apply a plant hormone source such as crushed bulb onion or Aloe gel to the cut end, because auxins stimulate roots to form

In a rooting test, four hibiscus cuttings were treated with crushed bulb onion, Aloe gel, cinnamon powder, or water only. Why is the cutting treated with water only included?

- A** It shows which cutting will die first
- B** It is the best treatment because water is the only thing roots need
- C** It makes sure all the cuttings get the same amount of onion
- D** It is the control, used to compare how many roots form with no rooting substance added

A farmer plants healthy-looking bean seeds in warm, moist soil with enough air. Most germinate, but a few do not even after many days. What is the most likely reason for the few that fail?

- A** The seeds are dormant, for example with a hard seed coat, so germination has not been triggered
- B** There is no water in the soil
- C** Beans do not have any hormones
- D** The seeds are monocots, so they cannot germinate

Which statement correctly describes germination in maize and in common bean?

- A** Both are epigeal, because all seeds push their cotyledons above the ground
- B** Maize is hypogeal, with its cotyledon staying underground, while bean is epigeal, with its cotyledons raised above the ground
- C** Both are hypogeal, because seeds always germinate underground
- D** Maize is epigeal and bean is hypogeal, because maize has two cotyledons

In a class test, 10 soaked bean seeds and 10 dry bean seeds were kept under the same warm conditions. 8 of the soaked seeds germinated and 3 of the dry seeds germinated. What percentage of the soaked seeds germinated?

A 50%

B 30%

C 80%

D 8%

A farmer in Kitale plants hybrid maize seeds in dry soil and they fail to germinate. What is the best advice, and why?

- A** Plant the seeds much deeper, because seeds do not need water to germinate
- B** Add hormones to the dry soil, because hormones make seeds grow without water
- C** Use bigger seeds, because only large seeds need water
- D** Plant when the soil is moist, because water activates enzymes and gibberellins that break down stored food for the embryo

A learner writes: 'Then I thought hormones feed a cutting so it can grow. Now I know...' Which ending is correct?

- A** hormones are food stores in the cotyledons that give the plant energy
- B** hormones are only found in animals, so plants use fertiliser instead
- C** hormones are needed in large amounts, like water, to make plants grow
- D** hormones are chemical messengers made in small amounts that control growth, for example auxins stimulate root formation and apical dominance

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).